Spectral-Spatial-Temporal Domain Adaptation with Learnable Sinc Filters, Dynamic Graph Modeling, and Bidirectional GRU for Subject-Independent Motor-Imagery EEG Classification

Bevara Ashok Varma
Department of Computer Science and Systems Engineering,
Andhra University(A),
Visakhapatnam, Andhra Pradesh.
bevaraashokvarma@gmail.com

Prof. V. Valli Kumari
Department of Computer Science and Systems Engineering,
Andhra University College of Engineering (A),
Visakhapatnam, Andhra Pradesh.
vallikumari@andhrauniversity.edu.in

Abstract— Subject independent motor-imagery (MI) electroencephalography (EEG) decoding is challenging due to the large inter-subject variability in terms of spatial organization, spectral and temporal characteristics of the signals. In this work we present an integrated spectral-spatial-temporal domain adaptation framework which tackles these issues in end-to-end fashion. First, each trial is normalized channel-wise and fed to a learnable Sinc filter bank which captures task relevant frequency responses directly from the waveforms. These representations are then used to form sample specific channel relationships using a dynamic graph neural network, thus enabling the model to capture interactions between the 22 EEG electrodes. The representations are pooled across frequencies and modeled using a bidirectional gated recurrent unit followed by a squeeze-and-excitation based recalibration. The final 64 dimensional representations are used simultaneously for classification of left/right fist movement intent, subject-domain prediction via a gradient reversal layer, and a 128 dimensional supervised contrastive projection head. At inference time, the learned batch normalizations are applied to the target recordings to reduce domain shifts without needing the targets' labels. The whole framework was assessed using the PhysioNet EEG Motor Movement/Imagery Dataset, runs 4, 8, and 12, with data from 109 subjects being available. Ten randomly selected target subjects were evaluated in a leave-one-target-out paradigm, where each test fold used 45 trials and 99 source subjects for training. Across all held out targets, the framework achieved 80.44 ± 8.18% accuracy, 0.6089 ± 0.1635 Cohen's kappa, and an aggregate binary ROC-AUC of 0.8964. We hypothesize that jointly modeling frequency selective responses, electrode relations, temporal context and domain invariance can lead to calibration-light MI BCI systems.

Keywords— brain-computer interface, motor imagery, electroencephalography, subject-independent EEG, domain adaptation, Sinc filters, graph neural network, bidirectional GRU, supervised contrastive learning, adaptive batch normalization.

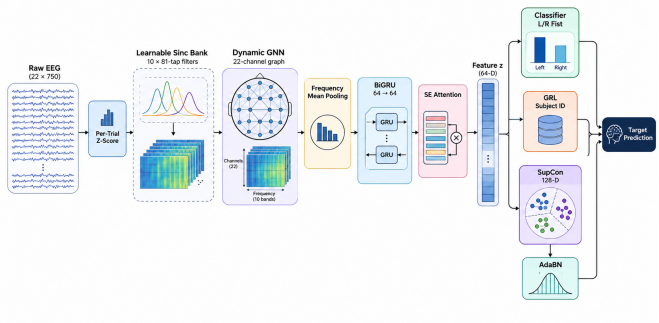


Fig. 1. Overview of the proposed spectral-spatial-temporal domain adaptation framework.

I. INTRODUCTION

Motor-imagery brain-computer interfaces (BCIs) interpret their user's intention from non-muscular voluntary brain activity. Imagining a left or right hand movement is a common paradigm, since such motor imagery produces readily-decodable sensorimotor rhythms in the form of EEG signals [1]. These systems have been explored as a means of communication, assistive technology, rehabilitation, and direct control over external devices. However, the practical utility of MI BCIs depends on the ability of a decoder trained from one user to generalize to a new user with little or no calibration.

That is the challenge that subject-independent EEG classification presents. The same imagined movement can manifest itself in significantly different amplitudes, frequencies, channel correlations, and time structures for each subject. Common spatial patterns (CSP) and other spatial filtering techniques continue to be popular in the literature, since they are effective at capturing discriminative variance structure. Nonetheless, these filters can be sensitive to noise, recording conditions, and the subject used to estimate them [2], [3]. Variants using filter-banks pay particular attention to the discriminative power of frequency structure in MI EEG, especially over the sensorimotor range [4], [5]. These points suggest the need for models that learn representations of both the spectrum and spatial structure, as well as the temporal structure of MI EEG.

Deep learning has enabled more flexible end-to-end decoding of EEG. Architectures such as EEGNet, DeepConvNet, and recurrent-convolutional designs have demonstrated that deep learning can learn useful representations of minimally-processed signals [6]-[8]. However, conventional classifiers still benefit from subject-specific clues that are predictive for the source population but not generalizable for a new person. Domain-adversarial training can discourage the retention of subject-specific clues, by suppressing the information in the feature extractor that makes target subjects easy to distinguish from the source [9]. In a similar vein, supervised contrastive learning encourages the formation of tighter clusters of samples with the same task label, which becomes attractive when inter-subject variability is large [10].

The architecture proposed in this paper combines these concepts with two components of signal-processing tailored to EEG. First, the front end uses parameterized Sinc band-pass filters, inspired by the compact, interpretable design of SincNet [11], to learn low-level spectral responses. Second, dynamic graphs are constructed for each sample, from the learned spectral descriptors, instead of treating the 22 channels as independent rows of a matrix. This enables the spatial representation to change depending on the observed EEG pattern, and retains the intuition that electrodes measuring related cortical activity should not be modeled independently [12].

Temporal structure is captured via a bidirectional GRU (BiGRU). The recurrent unit receives the graph-enhanced sequence, and learns temporal relationships in both directions. A squeeze-and-excitation block recalibrates the resulting channels, before being used by three learning objectives: task classification, subject-domain prediction (with a gradient reversal layer (GRL)), and supervised contrastive

learning. Finally, adaptive batch normalization (AdaBN) is performed on the target recordings, to update the normalization statistics before prediction, in line with the general principle that domain-specific activation statistics can be adapted without changing learned parameters [13].

The main contributions of this work are therefore four-fold: first, an end-to-end spectral-spatial-temporal architecture is presented, that learns frequency responses, electrode relationships, and temporal context jointly. Second, the model incorporates dynamic graph propagation via BiGRU and channel attention, to obtain a compact 64-dimensional latent representation. Third, domain alignment is enforced at both the subject-classification and representation-structure levels, using GRL and supervised contrastive learning. Fourth, the framework is evaluated using a cross-subject protocol on the PhysioNet EEG Motor Movement/Imagery Dataset, with all performance numbers reported directly from the completed experimental notebook.

II. RELATED WORK

A. Subject-Independent MI EEG Decoding

Subject dependent MI systems can benefit from a high performance due to the sharing of subject-specific statistics in the training and test recordings. This advantage is turned into a disadvantage if the model has to function for an unseen user. Kwon et al. investigated the subject independent MI decoding using a deep convolutional architecture and demonstrated that learning robust spectral-spatial representations can facilitate the cross-subject generalization [8]. Zhang et al. employed filter-bank processing, spatial features, recurrence and discriminative feature learning for the BCI Competition IV 2a dataset [14]. Their study is of particular interest in this context due to the emphasis on reducing intra-class variability, while preserving the discriminative information across subjects.

B. Frequency-Selective Learning

Fixed filter banks are still widely used in EEG pipelines, especially when prior information about sensorimotor rhythms is available. Sinc-based filters are an attractive alternative in which the lower and upper cutoff frequencies can be learned while forcing the filter to maintain a band-pass shape [11]. In the current model, ten learnable Sinc filters are applied independently to each EEG channel. This results in ten spectral views of the same trial and provides the next graph module with frequency-aware channel descriptors.

C. Graph Representations and Domain Adaptation

Graph convolutional networks enable modeling relational structures that are defined over connected entities rather than fixed Euclidean neighborhoods [12]. For EEG, electrodes form an informative graph because their physical arrangement and functional interactions cannot be captured fully by a 1D ordering of channels. Our graph is dynamic: the adjacency matrix is generated from the current sample rather than fixed across trials.

For domain shift, gradient reversal ensures that the feature extractor can be simultaneously informative for the classification objective and uninformative about the identity of the source subject [9]. Supervised contrastive learning adds to this objective by pulling together samples of the same class and pushing apart samples of other classes in an embedding space [10]. AdaBN provides a lightweight final alignment step by replacing statistics from source-domain normalization layers with target data estimates [13].

III. DATASET AND PREPROCESSING

A. PhysioNet EEG Motor Movement/Imagery Dataset

Experiments are based on the EEG Motor Movement/Imagery Dataset provided by PhysioNet [15], [16]. This database comprises 64-channel EEG recordings obtained using the BCI2000 instrumentation framework during motor-imagery and motor tasks. The experimental notebook utilizes the entire pool of 109 available subjects within the database, while the publication evaluation mode selects ten target subjects randomly: 4, 15, 23, 29, 31, 42, 55, 71, 82, and 95. Only runs 4, 8, and 12 are used in accordance with the required unilateral hand motor-imagery setting for the binary left/right experiment.

The first 22 channels are retained to ensure consistent inter-trial sensor arrangement. Signals are resampled at 250 Hz. Each selected event provides a 3.0-s analysis window between 0.5 s to 3.5 s relative to the detected motor-imagery event: 750 samples per channel are used as a fixed-size model input in the form of $\mathbf{X} \in \mathbb{R}^{22\times750}$.

B. Trial Normalization

As a preprocessing step before training, to reduce the amplitude offsets and differences in channel-wise scale, the data are normalized independently for every trial and every channel. The normalized sample would thus be

$$\tilde{x}_{c,t} = \frac{x_{c,t} - \mu_c}{\sigma_c + 10^{-6}}, . \quad (1)$$

Here, μ and σ are computed over time within that trial to prevent one or a few channels with particularly large absolute amplitudes from overwhelming the others during learning. The implementation used here matches what was done in the completed notebook.

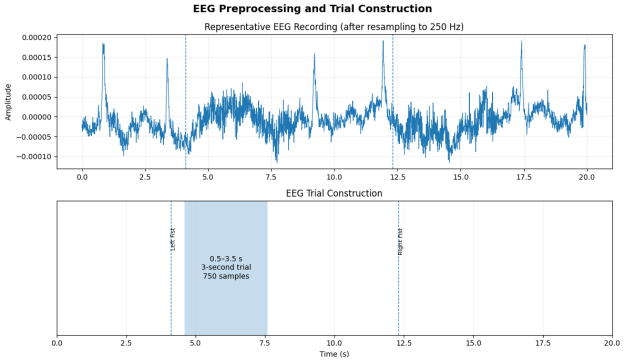


Fig. 2. EEG preprocessing and trial construction recorded by the experiment. The signal is resampled to 250 Hz and a 0.5–3.5 s window is extracted from the relevant event, giving 750 samples for each channel.

The notebook identified event labels from the recording annotations rather than assuming that the same numeric event labels would be used for all files. The events corresponding to the left and right fist imagery were then relabeled as 0 and 1, respectively.

IV. PROPOSED SPECTRAL-SPATIAL-TEMPORAL ADAPTATION FRAMEWORK

A. Overview

The processing pipeline is shown in Fig. 1. The architecture is partitioned so that each block targets a single aspect of the EEG problem. The Sinc filter bank encodes spectral perspectives, the dynamic graph learns spatial interactions, frequency pooling unifies the spectral branches, the BiGRU learns temporal context, and SE attention highlights informative latent channels. The resulting feature vector is shared by the task classifier and the adaptation objectives.

B. Learnable Sinc Filter Bank

A standard temporal convolution learns all coefficients of all kernels. In contrast, our first layer only learns two cutoff frequencies for each filter. For filter k, the impulse response is defined as difference of two low-pass Sinc functions:

$$w_k[n] = 2\frac{f_{2,k}}{F_s}\mathrm{sinc}\left(2\frac{f_{2,k}}{F_s}n\right) - 2\frac{f_{1,k}}{F_s}\mathrm{sinc}\left(2\frac{f_{1,k}}{F_s}n\right). \quad (2)$$

The implementation uses ten filters, an 81-tap kernel, and Fs = 250 Hz. Same bank is applied channel-wise, producing B ×10 ×22 ×750. Since the filter boundaries are learned during the

optimization process, the model can shift the spectral emphasis towards bands that are relevant for the decoding task.

This design preserves the explicitness of the frequency axis up to the graph stage. Instead of collapsing all the spectral information at once, the model can let each frequency view affect the learned channel relationships.

C. Dynamic Graph Neural Network

The graph module processes the 22 EEG channels on a graph structure, where each channel is a node. For all channels, the channel's signals are aggregated over time for each sample and frequency band. This is used as a compact descriptor for the node. Query- and key-projections are performed to map the node descriptors to a common space. The scaled dot product between query and key-projections is used to compute a dense attention matrix. This matrix is then normalized using a softmax and has an identity matrix added to it to retain self-connections, which is subsequently degree normalized to obtain a new graph adjacency matrix. This adjacency matrix is used to propagate the band-specific channel signals from the channels to their neighbors. The result of this propagation is then projected from the size of 22 channels to 64 features with a linear transformation and an ELU nonlinearity. In contrast to the anatomical graph, the adjacency matrix for the learned graph is not fixed, but predicted separately for each input sample, and thus can model different connectivity patterns per sample. The graph module is inspired by graph neural networks in general, which operate on a graph structure to aggregate information from related nodes [12].

D. Frequency Mean Pooling and Temporal Modeling

After graph propagation, the ten frequency branches are averaged out to form a single 64-channel temporal representation. In this way, the explicit spectral dimension is reduced to the information learned by the filter bank. The temporal sequence is processed by a bidirectional GRU with 32 hidden units per direction, resulting in a 64-dimensional output at every time step. The two directions provide context for a time point from both sides, while keeping the width of the final feature as a constant.

GRUs are compact recurrent units, and the gated nature provides a good structure for learning representations of sequences [17]. For this particular experiment, the temporal encoder is specifically a BiGRU and the architecture does not include a state-space or Mamba module.

E. Squeeze-and-Excitation Attention

The temporal outputs to an SE attention block. A global mean over time gives a 64-dimensional channel descriptor, which is compressed to four units, expanded back to 64 units, and passed through a sigmoid to determine channel-wise weights that recalibrate each latent channel before the final temporal mean is taken. This follows the general idea of explicit channel reweighting introduced by squeeze-and-excitation networks [18].

F. Multi-Objective Domain Adaptation

Let $z \in \mathbb{R}^{64}$ be the final feature vector. Three heads are applied to this representation: a task-specific head, a domain-specific head, and a contrastive head. The task head consists of a BatchNorm1d layer and a linear classifier that predict the two MI classes. Domain head is a 64-to-32 fully connected network, followed by a 32-to-109 linear classifier that predicts the source-subject identity. We apply gradient reversal layer (GRL) before the domain classifier, so that the feature extractor learns the domain-invariant representation [9]. The contrastive head is a 64-to-128-to-128 projection MLP, followed by L2-normalization and temperature scaling with 0.07. The supervised contrastive loss brings together the examples of the same class and pushes away the examples of a different class [10].

We define our final loss as

$$L = L_{\text{cls}} + L_{\text{dom}} + 0.5L_{\text{SupCon}},$$

where the domain term is weighted by one. We applied cross-entropy loss for the classification task with label smoothing of 0.1. The logistic scheduler smoothly grows the gradient reversal layer coefficient from near-zero to one over the training period, thus, preventing the domain classifier from overpowering the feature extractor at the beginning of the training.

G. Target-Time Adaptive Batch Normalization

Before target inference, the model updates the running statistics of its BatchNorm layers using the target recordings, but without target labels. The implementation then resets the running statistics, does a no-gradient target forward pass with disabled gradient reversal layer, re-enables the default momentum, and switches the model back to evaluation mode. This follows the principle of AdaBN that normalization statistics can encode the domain specific distribution information [13]. Since the target recordings are used to update the statistics, the protocol is best described as an unsupervised test-time adaptation rather than a pure zero-calibration evaluation.

V. EXPERIMENTAL PROTOCOL

A. Cross-Subject Evaluation

The publication evaluation mode performs ten target folds over the curated subjects. For each fold, the target subject is excluded from source training, and the remaining available subjects are sampled to form a source pool of 99 subjects. Each target contributes 45 trials (15 trials from each of the runs 4, 8, and 12) to the source pool. The ten folds therefore provide 450 target trials for final aggregate evaluation.

The source datasets contain small variations in trial availability as the recorded events are not identical for every subject and run. For example, the first target fold contains 4,468 source trials and 45 target trials. These values are taken directly from the executed notebook.

B. Training Configuration

The model is trained for 100 epochs using batches of 64 trials. We use AdamW with an initial learning rate of 10^{-3} and weight decay of 10^{-4}. The learning rate is decayed towards 1e-5 using a cosine annealing scheduler. We clip gradients at a maximum norm of 1.0. We set the domain loss weight to 1.0, and the SupCon weight to 0.5, and use a temperature of 0.07 for SupCon. The model uses 10 Sinc filters of length 81, and a 64-dimensional latent space. The completed run used the MPS backend on an Apple Silicon system.

Setting	Value
Available subjects	109
Source subjects / fold	99
Target folds	10
Runs	4, 8, 12
Classes	Left Fist, Right Fist
Channels	22
Sampling rate	250 Hz
Window	0.5–3.5 s
Samples / trial	750
Sinc filters / kernel	10 / 81 taps
Latent feature size	64
BiGRU hidden size	32 / direction
SupCon projection	64–128–128
Batch / epochs	64 / 100

Optimizer	AdamW
LR / weight decay	10^-3 / 10^-4
Label smoothing	0.1
Domain weight / SupCon weight	1.0 / 0.5
SupCon temperature	0.07
Gradient clipping	1.0

TABLE I. PRINCIPAL EXPERIMENTAL AND ARCHITECTURAL SETTINGS.

VI. RESULTS

A. Overall Performance

The finished experiment has such metrics as a mean target accuracy of 80.44 ± 8.18% over ten held-out subjects, Cohen's kappa of 0.6089 ± 0.1635 on average, and an aggregate binary ROC-AUC of 0.8964. Thus, it can be seen that the mean accuracy is about 30.44% higher than the 50% binary chance level. The standard deviation suggests that the model is efficient enough for the chosen users but still depends on the subject.

Target	Accuracy (%)	Kappa	Correct
S004	84.44	0.6884	38/45
S015	86.67	0.7332	39/45
S023	73.33	0.4664	33/45
S029	80.00	0.6002	36/45
S031	84.44	0.6884	38/45
S042	75.56	0.5114	34/45
S055	95.56	0.9112	43/45
S071	84.44	0.6890	38/45
S082	75.56	0.5104	34/45
S095	64.44	0.2899	29/45

TABLE II. TARGET-SUBJECT PERFORMANCE FROM THE COMPLETED TEN-FOLD EXPERIMENT.

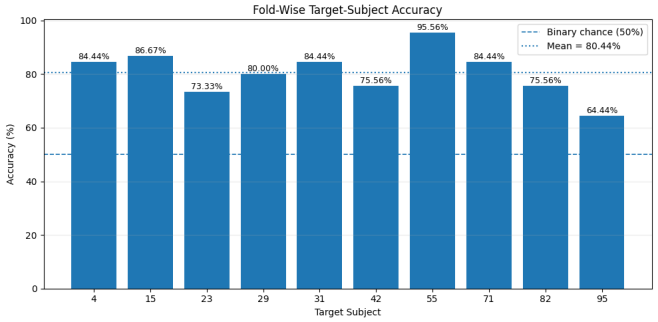


Fig. 3. Fold-wise target-subject accuracy. The mean performance is 80.44%, clearly above the 50% binary chance level, but the variation across users remains visible.

The fold-wise plot indicates that eight out of the ten selected targets demonstrated more than 75% accuracy, and one target approached 65%. The best result of 95.56% is achieved for the subject S055, for which the confusion matrix does not include any Right-to-Left errors. Meanwhile, the worst performance is observed for subject S095 yielding only 64.44% accuracy and a moderate kappa of 0.2899, indicating that even after adaptation, substantial inter-subject variability remains.

B. Training Behavior

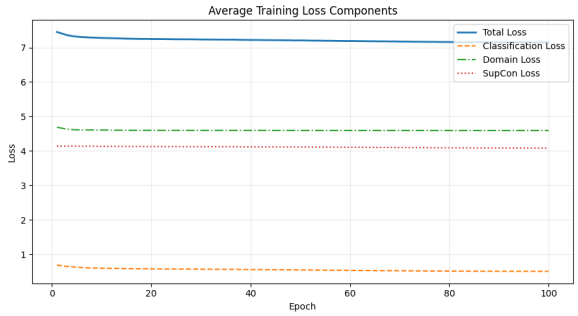


Fig. 4. Average training-loss components across the completed target folds. Classification loss decreases steadily, while domain and supervised-contrastive losses remain comparatively stable.

The average training-loss curves demonstrate a gradual reduction of the classification loss from approximately 0.7 at the beginning towards approximately 0.5 near epoch 100, while the domain loss remains close to 4.6 and the SupCon loss changes only slightly. Due to the combination of these terms with different weights in the total objective, the overall loss is still dominated by the domain and contrastive components. The trajectory of the loss is consistent with the simultaneous optimization of tasks and domain regularization.

C. Confusion Structure

The overall confusion matrix over the 450 target trials has 181 Left Fist trials correctly decoded by the model, 181 Right Fist trials decoded correctly, 45 Left-to-Right errors, and 43 Right-to-Left errors. This implies that the row-normalized recalls are 80.09% for the Left Fist and 80.80% for the Right Fist. The matrix is therefore approximately symmetric, which means the model is not significantly biased towards one class.

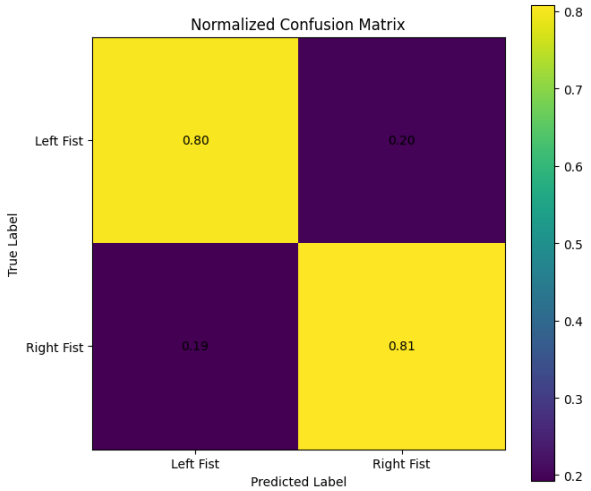


Fig. 5. Normalized aggregate confusion matrix for the two-class evaluation. Left Fist and Right Fist achieve similar row-normalized recall.

AOC value of the ROC curve is 0.8964. This implies that the learned decision scores maintain useful ranking information even for targets whose classification accuracy is below the overall means. ROC-AUC for a binary decoder is a way of separating score quality from the specific probability threshold used for the final decision on class membership.

D. ROC Analysis

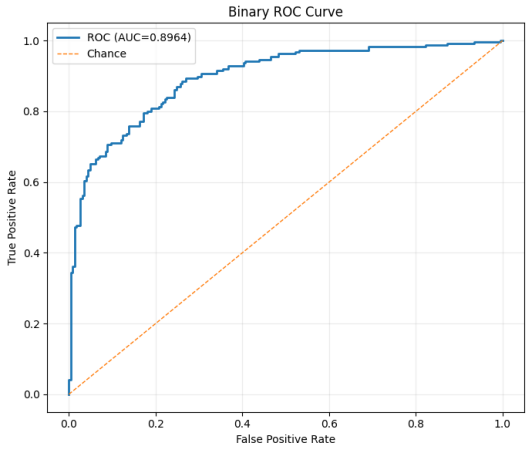


Fig. 6. Aggregate binary ROC curve across the ten target folds. The measured AUC is 0.8964.

E. Latent Representation

The 128 dimensional supervised-contrastive embedding was projected to two dimensions using t-SNE [19]. The resulting plot shows local organization for the two hand classes across the target recordings, albeit with some overlap. This qualitative behavior is in line with the intended role of the contrastive head: to preserve task structure while the domain objective suppresses subject-specific separability.

VII. DISCUSSION

The main finding from our experiments is that the problem is not so trivial as just picking a better classifier. The network has to process a signal which is inseparable in terms of spectral, spatial, and temporal information with subject identity. Our architecture addresses all three aspects which were sources of domain shift.

The learnable Sinc bank is necessary because it does not require the network to learn narrow-band filtering with unconstrained time-domain filters. The first layer is constrained to implement band-pass filtering and can be trained to shift center frequencies. This gives a compact representation in the spectral domain while retaining physical interpretability of the front end [11]. The dynamic graph construction step does not use a predefined graph but rather encodes subject-specific connections between channels which are then used by the graph convolution. Finally, the Temporal module has been kept simple: after encoding the spectral and spatial information, the 22-channel input is reduced to 64 latent channels. Therefore, to model temporal information, a simple BiGRU is sufficient, while additional convolutional layers would unnecessarily expand the number of parameters. The SE block can be seen as the final step of subject-specific feature selection, where certain channels are amplified while others are attenuated [18].

The domain adaptation branch is crucial because optimizing the task objective is insufficient for achieving subject-invariant features. The gradient reversal layer regularizes the feature extractor to prevent the subject classifier from performing well, while the contrastive objective uses the supervisory signals to guide the adaptation. The two regularization techniques are not equivalent: By design, the adversarial objective removes all subject-discriminatory features, while SupCon preserves class-discriminatory features. The two branches complement each other, which explains the synergy.

The final test-time adaptation step is a practical consideration: Since batch normalization statistics are part of the model forward pass, adapting them to the target distribution modifies the representation of the features without requiring parameter updates [13]. This adaptation only requires target-domain EEG recordings (without labels), making it more desirable from a deployment perspective. However, this also means that we are not evaluating the zero-shot performance; we need some unlabeled target examples to perform adaptation.

Compared to the subject-independent hybrid framework proposed by Zhang et al. [14], the main difference is that we have adapted the representation learned by the subject-dependent network rather than generating target-like EEG signals for domain adaptation. We are not directly comparing our method to existing approaches in terms of final metrics because our experiments differ in many respects: the task, preprocessing, target population, and evaluation protocol. Most importantly, our result is achieved with a different target-fold partitioning than the one used in the comparison paper. Therefore, our result of 80.44% should be considered as the performance of our implementation for this particular hold-out set.

A. Limitations

Several limitations must be acknowledged. First, the final publication evaluation uses ten curated target subjects rather than all 109 as independent targets, and thus the result should not be described as an exhaustive 109-fold cross-subject benchmark. Second, each fold takes 99 source subjects from the remaining pool, and so while the source population is large, it is not identical to an exhaustive all-source protocol. Third, the target recordings used for AdaBN are also the recordings that are subsequently evaluated for accuracy, though labels are not used during the adaptation pass. This is a valid unsupervised test-time adaptation design, but it should be noted explicitly when compared against methods which prohibit any access to test-domain samples before prediction. Fourth, the notebook does not contain ablation experiments which isolate the contribution of each component, and so the present paper does not claim that any single module is individually responsible for the final accuracy. Finally, the 8.18 percentage-point standard deviation and the 64.44% result for S095 shows that there remains substantial user-to-user variability.

VIII. CONCLUSION

In this paper we presented a complete spectral-spatial-temporal domain adaptation framework for subject-independent binary MI EEG classification. The system learns frequency responses with a ten-filter Sinc front end, a sample-adaptive graph of channel interactions, temporal dynamics captured with a BiGRU, latentchannel recalibration via SE attention, and domain-aware representation learning with GRL and supervised contrastive objectives, and uses target-time AdaBN to perform a lightweight distribution-alignment step prior to prediction.

On the completed ten-target evaluation from the experimental notebook, the framework achieved 80.44 ± 8.18% mean accuracy, 0.6089 ± 0.1635 mean Cohen's kappa, and 0.8964 ROC-AUC over 450 target trials. These results support the general hypothesis that subject-independent EEG decoding benefits most when spectral selection, spatial relations, temporal context, and domain mismatch are treated as integrated components rather than isolated preprocessing decisions. The next step is a controlled ablation and exhaustive subject-wise evaluation in which every component is tested separately and the role of target-time adaptation is quantified independently.

ACKNOWLEDGMENT

We acknowledge the public PhysioNet EEG Motor Movement/ Imagery Dataset and the open research infrastructure that makes reproducible EEG experimentation possible.

DATA AVAILABILITY

The experiments use the publicly available EEG Motor Movement/ Imagery Dataset that is distributed through PhysioNet. The full preprocessing, model definition, training procedure, evaluation protocol, and result logs are contained in the accompanying Jupyter notebook that has been used for this study.

REFERENCES

[1] G. Pfurtscheller and C. Neuper, "Motor imagery and direct brain-computer communication," Proc. IEEE, vol. 89, no. 7, pp. 1123–1134, 2001.

[2] H. Ramoser, J. Muller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," IEEE Trans. Rehabil. Eng., vol. 8, no. 4, pp. 441–446, 2000.

[3] F. Lotte and C. Guan, "Regularizing common spatial patterns to improve BCI designs: Unified theory and new algorithms," IEEE Trans. Biomed. Eng., vol. 58, no. 2, pp. 355–362, 2011.

[4] K. K. Ang, Z. Y. Chin, H. H. Zhang, and C. Guan, "Filter bank common spatial pattern (FBCSP) in brain-computer interface," in Proc. Int. Joint Conf. Neural Netw., 2008, pp. 2390–2397.

[5] S. Lemm, B. Blankertz, G. Curio, and K.-R. Muller, "Spatio-spectral filters for improving the classification of single trial EEG," IEEE Trans. Biomed. Eng., vol. 52, no. 9, pp. 1541–1548, 2005.

[6] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," J. Neural Eng., vol. 15, no. 5, p. 056013, 2018.

[7] R. T. Schirrmeister et al., "Deep learning with convolutional neural networks for EEG decoding and visualization," Hum. Brain Mapp., vol. 38, no. 11, pp. 5391–5420, 2017.

[8] O.-Y. Kwon, M.-H. Lee, C. Guan, and S.-W. Lee, "Subject-independent brain-computer interfaces based on deep convolutional neural networks," IEEE Trans. Neural Netw. Learn. Syst., vol. 31, no. 10, pp. 3839–3852, 2020.

[9] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in Proc. Int. Conf. Mach. Learn., vol. 37, 2015, pp. 1180–1189.

[10] P. Khosla et al., "Supervised contrastive learning," in Adv. Neural Inf. Process. Syst., vol. 33, 2020.

[11] M. Ravanelli and Y. Bengio, "Speaker recognition from raw waveform with SincNet," in Proc. IEEE Spoken Lang. Technol. Workshop, 2018, pp. 1021–1028.

[12] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in Proc. Int. Conf. Learn. Represent., 2017.

[13] Y. Li, N. Wang, J. Shi, J. Liu, and X. Hou, "Revisiting batch normalization for practical domain adaptation," in Proc. Int. Conf. Learn. Represent., 2017.

[14] H. Zhang, H. Ji, J. Yu, J. Li, L. Jin, L. Liu, Z. Bai, and C. Ye, "Subject-independent EEG classification based on a hybrid neural network," Front. Neurosci., vol. 17, p. 1124089, 2023, doi: 10.3389/fnins.2023.1124089.

[15] G. Schalk, "EEG Motor Movement/Imagery Dataset," PhysioNet, version 1.0.0, 2009, doi: 10.13026/C28G6P.

[16] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," Circulation, vol. 101, no. 23, pp. e215–e220, 2000.

[17] K. Cho et al., "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in Proc. Conf. Empirical Methods Natural Lang. Process., 2014, pp. 1724–1734.

[18] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2018, pp. 7132–7141.

[19] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," J. Mach. Learn. Res., vol. 9, pp. 2579–2605, 2008.

[20] J. Yang, S. Yao, and J. Wang, "Deep fusion feature learning network for MI-EEG classification," IEEE Access, vol. 6, pp. 79050–79059, 2018.

[21]Tang, Xianlun & Zhang, Jing & Qi, Yidan & Liu, Ke & Li, Rui & Wang, Huiming. (2023). A Spatial Filter Temporal Graph Convolutional Network for decoding motor imagery EEG signals. Expert Systems with Applications. 238. 121915. 10.1016/j.eswa.2023.121915.